Signed order count, predictable flow, and order-book imbalance

TSLA, 2019

Conditional impact, chronological prediction, and depth-aware replay

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This analysis began in the course project *Liquidity Games in High-Frequency Markets*, supervised by Anastasia Bugaenko of Capital Fund Management. The course presentation was prepared by L. Cohen, R. Darwish, A. Pariente, H. Rohrbach, and R. Sithisak. I later rewrote the supervised analysis and prepared this report in a separate repository.

Abstract

The analysis reconstructs 4,547,078 visible market orders from 249 included sessions among 252 delivered TSLA file pairs in 2019. LOBSTER messages are aligned with the book state immediately before each event, and fills with the same timestamp and initiating side are grouped. The main experiment relates signed volume and signed order count in a 10-order window to the mid-price change over that window. Since the regressors are observed during the window, the exercise estimates conditional impact rather than a pre-trade forecast.

The fixed late-year dates form an untrimmed holdout of 102,468 windows. Signed-volume OLS reaches $R^2 = 0.118$. Nonlinear volume terms raise the score to 0.298, and raw signed order count raises it to 0.359. Further count transforms change little: the full score is 0.360. Adding raw count to the nonlinear volume model reduces mean squared error by 8.7%. Resampling complete test dates gives a 95 percent interval of 8.0% to 9.5% for that incremental reduction. The full model's total reduction relative to linear signed volume is 27.4%.

Separately, daily DFA1 gives a median order-sign exponent of 0.699 against a within-date permutation-null median of 0.501 ($p_{\mathrm{MC}} = 0.005$). This establishes serial dependence on the fixed scales, not long memory.

A fixed AR(50) of visible-order signs then reduces later-date sign MSE by 39.6% and shows that highly expected orders meet or exceed the initial opposite best quote less often than surprising orders. Expected orders are slightly larger, but the displayed depth opposite them increases much more, so their mean log size-to-depth ratio is lower. The same attribution survives a fixed date-by-side-by-half-hour control. This post-hoc result is an association, not a causal efficiency test.

Separately, a C++20 engine audits 38,516,432 included level-2 message/snapshot pairs without claiming full order priority. It classifies 28,544,808 transitions as exact and 9,971,375 as depth-censored, with no observable mismatch, unsupported event, or invalid snapshot. A forward-looking experiment then predicts the direction of the next same-session mid-price change from current queue imbalance and causal top-of-book OFI. The combined model reaches ROC-AUC 0.627 and reduces Brier score by 5.61% across all spreads. In the exploratory one-tick subset, the corresponding values are 0.752 and 18.24%. A nested comparison selects fixed OFI lookbacks using only earlier dates, but the selected windows have 2.09% and 5.71% more Brier error than the price-spell reset on the pooled and one-tick holdouts. A state-level marketable-markout gate subtracts the displayed half-spread and delays entry. At one tick, its top 10-percent training-confidence tail has a net markout of 0.399 basis points immediately and 0.302 at 10 microseconds, but the 100-microsecond interval crosses zero; the corresponding all-spread immediate markout is -1.254. A fixed 100-microsecond landmark then permits at most one signal per price spell. Its one-tick net markout falls from 0.107 immediately to 0.059 after 10 microseconds and -0.095 after 100 microseconds. In a monthly rolling-origin audit of those non-overlapping landmarks, the combined model reduces Brier error by 1.17% across all spreads and 5.55% at one tick over the final six calendar months, with positive reductions in every month. A depth-constrained gate then crosses the displayed book again at the next price move. The one-share, one-tick quoted round trip is -0.684 basis points $[-0.756, -0.619]$ before fees. Across the full 360-policy grid, the best observed quoted result remains negative at -0.615 basis points, with a selection-adjusted simultaneous interval $[-0.709, -0.520]$.

I also apply the finite-size scaling construction of Patzelt and Bouchaud (2018) to the same stock-year. The width and height scale regressions have in-sample R^2 values of 0.9958 and 0.9975. These fits use 28 estimated scale points and are unrelated to the chronological prediction score.

Source-data status. The original project access has ended and replacement files are unavailable. A shared policy declares the 3 delivered files that end before their scheduled close as source exclusions. The Python gate audits all 252 pairs, and the C++ validator applies the same rules to decoded sessions. The shared counts require exactly 249 included sessions with no undeclared coverage failure. Every reported artifact was regenerated on that universe; the study does not claim complete 2019 coverage or external replication.

Keywords: market impact, order-sign persistence, detrended fluctuation analysis, asymmetric liquidity, predictable order flow, marketable markout, quoted round trip, price-spell landmark, queue imbalance, order-flow imbalance, market microstructure, chronological validation, rolling-origin evaluation, order-book replay, finite-size scaling

1 Question

In the local benchmark of Kyle (1985), signed order flow and price change are connected by a linear slope. Empirical impact curves bend, change scale with the aggregation horizon, and reflect persistent trade signs (Bouchaud et al., 2018; Patzelt and Bouchaud, 2018). I ask four distinct questions for TSLA in 2019. First, does signed order count explain conditional price impact on later dates after signed volume has already been included? Second, do transaction signs persist on fixed within-session scales, and when prior signs make an aggressive order more predictable, does the liquidity displayed against that order differ? Third, do the current queues and order-flow changes observed since the last price move forecast the direction of the next move? Fourth, does that direction signal survive the displayed spread, an explicit delay between decision and entry, and a quoted marketable exit?

Signed volume alone explains 0.118 of the holdout variance. Fixed volume transforms explain 0.298. Adding raw signed count brings the score to 0.359, almost the same as the full specification at 0.360. Raw signed count provides the second, incremental gain after volume curvature. Extra transforms and ridge regression leave the reported score unchanged.

Both regressors and response cover the same event-time window. The date split tests whether a conditional mapping estimated earlier in the year transfers to later dates. It does not identify a causal effect because order flow and liquidity react to each other.

The primary holdout keeps every observation. I estimate uncertainty by resampling trading dates, then repeat the comparison over expanding test blocks, five window lengths, and a basis-point response. The earlier trimmed protocol is reported separately because it produced the original 0.24 to 0.38 comparison.

2 Relation to prior work

Kyle's model gives a local liquidity measure: a larger price response per unit of signed flow corresponds to a less deep market (Kyle, 1985). It does not imply that the empirical response must remain linear far from balanced flow. The evidence reviewed by Bouchaud et al. (2018) shows why impact is difficult to see at the trade level and why conditional averaging is central to measurement.

Patzelt and Bouchaud (2018) study volume and sign imbalance over several intraday horizons. Their conditional curves can be rescaled by a horizon-dependent width and height. The published claim is based on many instruments and periods. Here, the same functional form is fitted to one included TSLA stock-year. This within-sample application can assess its description of these sessions; it is not an external replication of the paper's cross-sectional evidence.

Lillo and Farmer (2004) use a linear autoregression of prior order signs, with a largest lag typically around 50, to condition the probability that an aggressive order exhausts the opposite best quote. They also study the log ratio of order size to opposite displayed depth; that ratio mixes two channels, aggressive-order size and passive depth. Their evidence and the asymmetric-liquidity interpretation of Farmer et al. (2006) motivate the audit below. Because depth varies through the day, I use the half-hour partition in Cont et al. (2014) as a fixed post-hoc control. The TSLA exercise changes the binning and uncertainty protocol and does not reproduce the papers' multi-stock evidence.

The course project followed Bugaenko (2019), which connects order-flow imbalance, local liquidity estimates, and supervised impact models. I retain the transaction reconstruction and scaling exercise, but rebuild the supervised evaluation around untrimmed chronological date splits. The model ablation isolates signed count as the useful extension of a nonlinear volume baseline.

Gould and Bonart (2015) study queue imbalance as a one-tick-ahead price predictor and emphasize its dependence on the spread regime. I use the displayed level-1 imbalance for a separate next-mid-price-direction experiment. The pooled sample is the primary comparison; the one-tick subset is reported as exploratory.

3 Data and transaction reconstruction

3.1 LOBSTER alignment

LOBSTER provides a message file and an order-book file for each session. Message row k is the event that moves the book from row $k-1$ to row k (LOBSTER, 2026). The pre-event best bid and ask for an execution on row k therefore come from order-book row $k-1$. The pipeline makes that shift before calculating the mid-price or spread. An execution on the first row of a session is discarded because its pre-event state is absent.

Prices in the source files are integers scaled by 10,000. For a visible execution, event type 4, LOBSTER's direction field identifies the resting order. The aggressor sign is its opposite: $+1$ for a buyer-initiated execution and -1 for a seller-initiated execution.

For the liquidity audit, a realized buy uses the pre-event best ask size and a realized sell uses the pre-event best bid size. When several fills share one timestamp and initiating side, their reconstructed order keeps the opposite depth immediately before the first fill.

3.2 Source coverage

LOBSTER filenames state the requested time range, but that range alone does not show that the delivered rows reach the close. The audit therefore finds the last message at or before the scheduled close and requires it to fall within 60 seconds. Nasdaq's 2019 notices set a 13:00 close on 3 July, 29 November, and 24 December; the other files are checked against 16:00.

249 of the 252 delivered sessions pass. The files for 9 January, 8 March, and 18 September end 22,827, 958, and 5,403 seconds early. The original project access has ended and replacement files are unavailable, so the shared analysis policy declares exactly those three dates as source exclusions. Both Python annual readers audit every delivered pair, include the other 249 sessions, and reject any additional incomplete input before writing annual output. The C++ validator consumes the same policy for decoded sessions and exact universe counts. The daily aggregate record is committed as `results/session_coverage.csv`; it contains no licensed row, event price, or order identifier. Rows after the scheduled close are counted in the audit and excluded from prepared sessions: 909 on 3 July, 301 on 29 November, and 774 on 24 December. The fixed policy preserves development through 6 August, selection from 7 August through 17 October, and test from 18 October, giving 148/50/51 nested dates and a 198/51 outer split.

A market order may consume several displayed orders. The source used here does not expose a parent market-order identifier, so visible fills with the same date, timestamp, and aggressor sign are grouped. Share size is summed, execution price is volume weighted, and the first pre-event book defines the starting state. This is a reconstruction rule, not investor identification.

3.3 Sample

Table 1 reports descriptive quantities, not model results. The median reconstructed order has 55 shares. The distribution is right-skewed: its 90th and 99th percentiles are 195 and 700 shares. The median quoted spread is 7.0 cents, and the share of buyer-initiated orders is 49.7%.

Table 1. Reconstructed visible-order sample. Daily quantiles are computed across 249 sessions; order-level quantiles use all reconstructed visible market orders.

Quantity	Value
Timestamp-aggregated visible market orders	4,547,078
Median orders per session	16,049
10th to 90th percentile, orders per session	10,839 to 29,935
Median order size, shares	55
90th and 99th percentile order size, shares	195, 700
Median and 90th percentile spread, cents	7.0, 15.0
Buyer-initiated fraction	49.7%
Non-overlapping 10-order windows	454,576
Standard deviation of 10-order impact, cents	12.46
Zero-impact share of 10-order windows	3.0%

The raw and derived row-level files remain outside the repository because the LOBSTER license does not permit redistribution. Committed CSV and JSON files contain aggregate curves, scores, and sample summaries only.

3.4 A separate scaling table

The supervised experiment uses visible executions because their aggressor side is given by the resting-order direction. The scaling exercise also uses hidden executions, event type 5, to match the broader transaction definition in Patzelt and Bouchaud (2018). Their signs are inferred from execution price relative to the pre-event mid-price. Midpoint executions have no inferred side and are removed. After timestamp grouping, this table contains 5,845,437 transactions. The prediction and scaling exercises therefore use different transaction tables.

4 Measurement

Consider a same-session window beginning at reconstructed market order t and containing T signed orders. With aggressor sign ϵ_i, share size v_i, and pre-event mid-price m_i, define

$$\Delta V_{t,T} = \sum_{i=t}^{t+T-1} \epsilon_i v_i, \tag{1}$$

$$\Delta E_{t,T} = \sum_{i=t}^{t+T-1} \epsilon_i, \tag{2}$$

$$Y_{t,T} = 100\,(m_{t+T} - m_t)\,. \tag{3}$$

Here, Y is measured in cents. Windows are non-overlapping within each horizon and never cross a session boundary. A T-order window needs the mid-price at its right edge, so the final incomplete block of each session is discarded.

The count imbalance ΔE distinguishes flow patterns that signed volume alone merges. One large buy and several smaller buys can generate comparable ΔV but different ΔE. That difference may reflect order splitting, replenishment, or state-dependent order size. The regression cannot separate those mechanisms, but it can test whether the count adds conditional information on dates not used for estimation.

Figure 1 displays the raw conditional mean at four horizons. Quantile bins are used only for the figure. The outer bins contain wide ranges of volume, while the central bins reveal a steep response around balanced flow. The flattening in both tails is visible at every horizon. A single global linear coefficient cannot capture this shape.

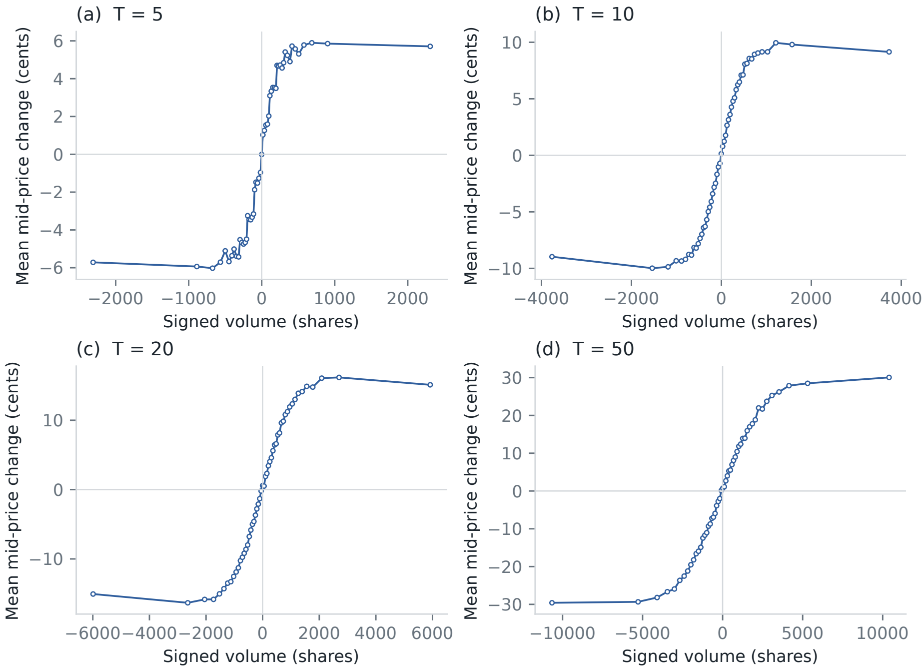

Figure 1. Mean mid-price change conditional on signed volume. Each point is a mean within one of 51 signed-volume quantiles. All 249 dates enter; the bins are not used for model fitting.

A local Kyle slope is also estimated within the central 35 percent of absolute signed volume. At $T = 10$, the fitted value is 0.01837 cents per share. Across $T \in \{5, 10, 20, 50, 100\}$, the fitted slopes decay approximately as $T^{-0.315}$. This is a descriptive liquidity estimate, not a trade-level execution-cost model.

5 Evaluation design

5.1 Chronological holdout

The main $T = 10$ sample contains 454,576 windows across 249 dates. Dates through 17 October form the training partition and dates from 18 October form the holdout. This fixed calendar boundary was retained after the source exclusions, so no inspected test date moved into training. It contains 198 included training dates and 51 holdout dates. Table 2 gives the exact boundary. No row is removed from either side of the split.

Table 2. Primary chronological partition.

Partition	First date	Last date	Windows
Train	2019-01-02	2019-10-17	352,108
Holdout	2019-10-18	2019-12-31	102,468

The four nested OLS specifications are:

$$\begin{aligned}
&\mathcal{M}_1 : \Delta V,\\
&\mathcal{M}_2 : \Delta V, |\Delta V|, \operatorname{sgn}(\Delta V)\sqrt{|\Delta V|}, \operatorname{sgn}(\Delta V)\log(1+|\Delta V|),\\
&\mathcal{M}_3 : \mathcal{M}_2, \Delta E,\\
&\mathcal{M}_4 : \mathcal{M}_3, |\Delta E|, \operatorname{sgn}(\Delta E)\sqrt{|\Delta E|}.
\end{aligned}$$

The transforms are fixed before estimation. An intercept is included. Ridge regression with standardized $\mathcal{M}_4$ features is retained as a numerical check, but its predictions are indistinguishable from OLS at the shown precision.

Scores are calculated only on the holdout:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2, \tag{4}$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{5}$$

Mean absolute error and sign accuracy are reported as secondary diagnostics. Sign accuracy omits a window when either the realized or fitted response is exactly zero.

5.2 Uncertainty and stability checks

Event-time windows from one session share market conditions. Treating all windows as independent would understate uncertainty. The bootstrap therefore resamples the 51 complete holdout dates with replacement. For each of 10,000 repetitions, scores are recomputed from the selected daily blocks. The interval is the 2.5th to 97.5th percentile of the bootstrap distribution.

Five additional temporal splits begin with 40 percent of the dates for training. Each subsequent block is tested once, then added to the expanding training set. Horizon checks retain the 18 October boundary for $T = 5, 10, 20, 50, 100$. Finally, the response is replaced by $10{,}000\log(m_{t+T}/m_t)$ to check that the result is not an artifact of measuring a changing stock price in cents.

6 Holdout results

6.1 Feature ablation

Table 3 separates curvature from count information. The linear volume baseline has $R^2 = 0.118$. Fixed volume transforms account for the first increase, to 0.298. Adding only raw signed count reaches 0.359. The two additional count transforms change the third decimal place by less than one point. Raw signed count accounts for the useful part of the second increase.

Table 3. Full chronological holdout, $T = 10$. Errors are measured in cents.

Specification	R^2	MSE	MAE	Sign accuracy
Signed volume	0.118	147.23	8.64	77.0%
Volume transforms	0.298	117.16	7.79	76.9%
Volume transforms plus raw signed count	0.359	106.93	7.48	78.5%
Volume and count transforms	0.360	106.86	7.47	78.5%

Figure 2 compares fitted and realized means in 20 holdout bins. The nonlinear volume-only model still leaves a systematic count residual. Adding raw signed count follows the binned means more closely, although individual windows remain noisy.

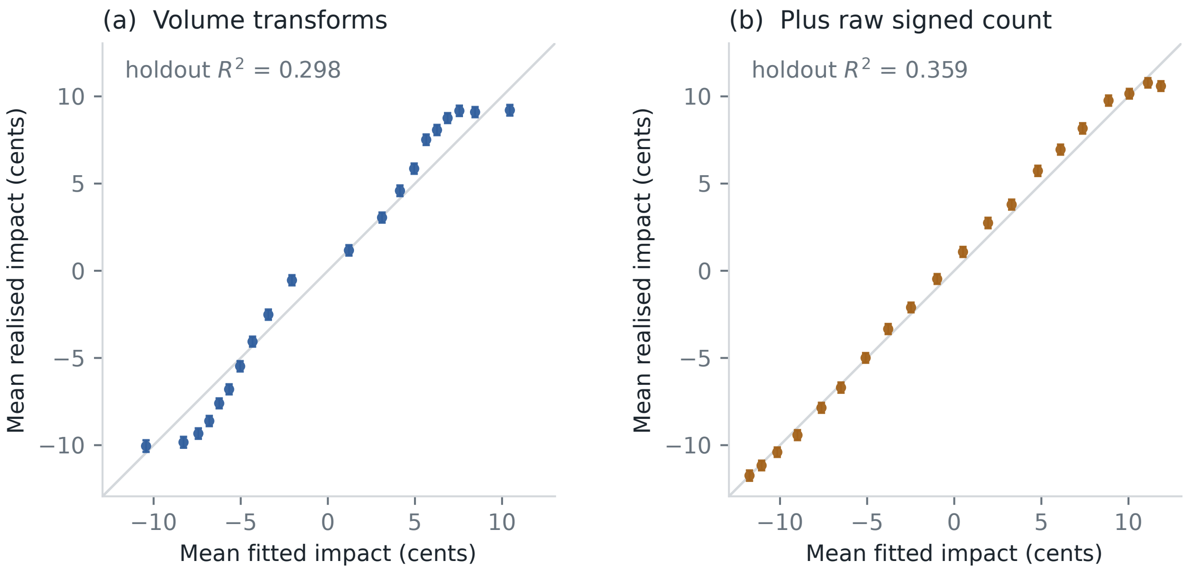

Figure 2. Holdout calibration by fitted-value quantile. Points show mean fitted and realized impact in 20 equally populated bins. Error bars are 1.96 observation-level standard errors and are included only as a visual guide; the formal score interval is clustered by date.

Figure 3 groups the holdout residual of the volume model by signed count. The mean moves from roughly −4 cents for ten sells to more than 3 cents for strong buy counts. This monotone residual pattern is the signal captured by ΔE.

Figure 3. Holdout residual from the volume-transform model, grouped by signed order count. Points are equal-weighted means across dates. The band is 1.96 standard errors of the date-level means.

Adding raw count lowers MSE from 117.16 to 106.93 cents squared, a relative reduction of 8.7%. The day-clustered 95 percent interval is [8.0%, 9.5%]. The corresponding increase in R^2 is 0.061, with interval [0.056, 0.066]. None of the 10,000 bootstrap draws gives a non-positive improvement. This interval quantifies sampling variation across the 51 late-year dates. It does not cover model-selection or cross-year uncertainty. For context, the full count-transform model lowers MSE by 27.4% relative to linear signed volume; that is a total ablation gap, not the effect of count alone.

6.2 Time and horizon checks

The raw-count specification outperforms the nonlinear volume model in every expanding test block in Figure 4. Its R^2 ranges from about 0.34 to 0.38, while the volume-only score ranges from about 0.29 to 0.31. The gap is present before the final December block.

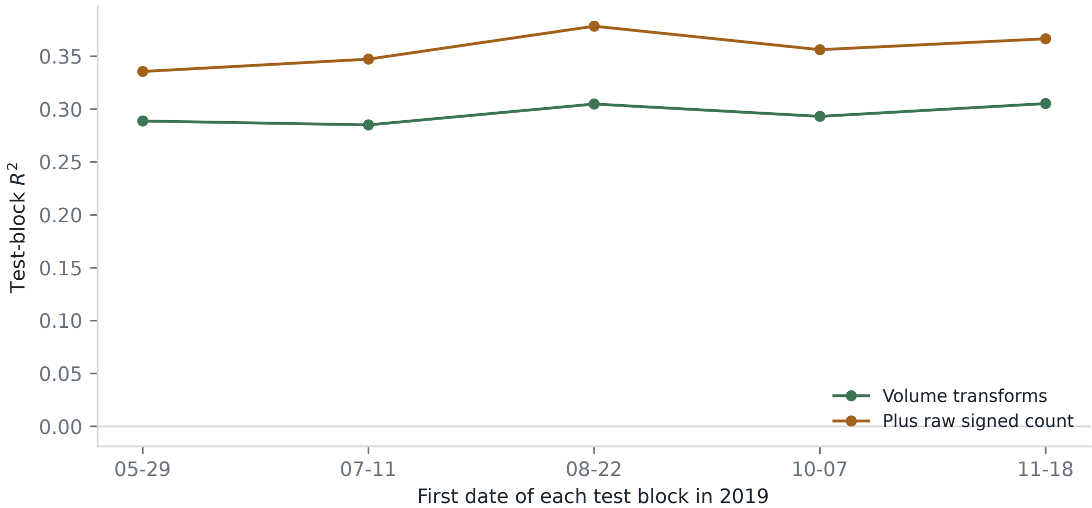

Figure 4. Five expanding-window evaluations. Each point is scored on the contiguous block beginning at the date shown. Earlier test blocks are added to the training set before the next fit.

The same ordering holds from 5 to 100 orders (Figure 5). At $T = 5$, R^2 rises from 0.237 to 0.282. At $T = 100$, it rises from 0.375 to 0.483. Longer windows average more idiosyncratic trade-level noise and contain a wider count range, so higher scores at larger T should not be read as better short-horizon forecasts.

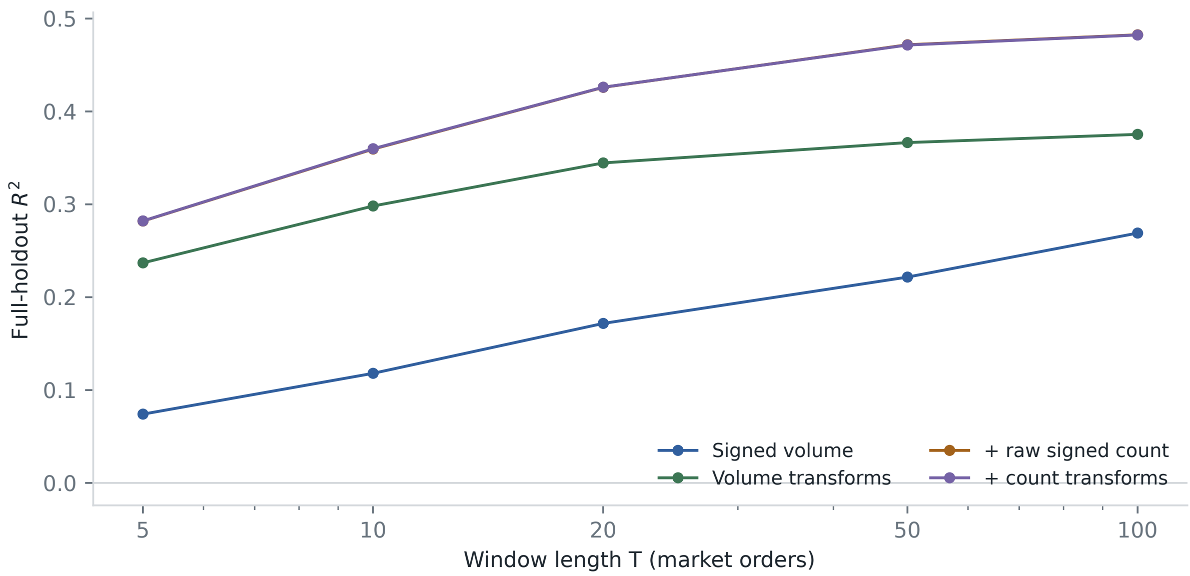

Figure 5. Full-holdout R^2 across event-time horizons. The raw count term accounts for nearly all of the gap between the volume-transform and full specifications.

The holdout result is also present within each calendar month. The augmented R^2 is 0.329 in the October portion, 0.378 in November, and 0.363 in December. With log return in basis points as the response, the scores are 0.281 and 0.307, and MSE falls by 3.7%. Price normalization reduces the size of the gain but does not remove it.

6.3 Why the previous score was 0.24 to 0.38

The earlier course protocol removed observations beyond three standard deviations in both signed volume and realized impact. Re-estimating those bounds on training dates avoids using the test distribution to set the cutoff, but applying the impact cutoff to the test still selects observations by the realized response. Under that trimmed comparison, the baseline has $R^2 = 0.241$ and the augmented model has $R^2 = 0.383$. MSE falls by 18.7%.

The trimmed scores describe the retained central sample. Selection on realized test impact makes them unsuitable as the primary evaluation. The cutoff raises the baseline much more than the count model, explaining the difference between 0.24 to 0.38 and the untrimmed 0.12 to 0.36 comparison.

7 Depth-aware L2 replay and next-price direction

7.1 Finite-depth transition audit

A finite-depth LOBSTER file is not a historical level-3 feed. Its message rows describe events that update the requested price range (LOBSTER, 2026). When a displayed level is depleted, a new tail level can enter from outside depth 2. Orders that formed that tail while it was unobserved are absent, so reconstructing their FIFO queues would invent state.

The C++20 audit seeds each included session from its first authoritative snapshot. It applies each later observable event to the previous two levels and compares the result with the supplied post-event snapshot. A transition is *exact* when every resulting level was known. It is *depth-censored* when depletion exposes an unobserved tail: the surviving prefix is checked and replay then resumes from the authoritative snapshot. Prices, sizes, and timestamps remain fixed-point integers.

Table 4. Included-session transition audit. The first row of each of the 249 sessions seeds replay and is not classified as a transition.

Quantity	Value
Message/snapshot pairs	38,516,432
Fully determined transitions	28,544,808 (74.11%)
Depth-censored transitions	9,971,375 (25.89%)
Observable mismatches	0
Unsupported events / invalid snapshots	0 / 0

Zero mismatches means that all observable prices and sizes agree under this contract. It does not turn level-2 data into an exchange matching engine, validate an unobserved queue, or establish execution priority.

7.2 Benchmark protocol

The benchmark separates complete CSV decoding from replay over the resident 64-byte event records. The measured replay audits transitions, checks snapshot invariants, and maintains a checksum. The aggregate signal exports below run separately after those measured passes.

Table 5. Single-threaded included-session benchmark on an Apple M4 Pro, 48 GB, compiled for arm64 with Apple Clang 17 in Release mode.

Phase	Run-average ns/event	Million events/s
CSV decode, median of 3 passes	115.19	8.68
Resident replay, median of 11 passes	9.22	108.46

Two unmeasured replay warm-ups precede the eleven passes. Their p95 run-average is 9.51 ns/event. The operating-system page cache is not flushed between decode passes: the first recorded pass costs 277.33 ns/event. Each timing divides a complete pass by 38,516,432. These are throughput measurements, not individual-event tail latency or wire-to-wire trading latency.

7.3 Causal queue and order-flow signals

For the displayed best queues q_t^b and q_t^a, define

$$I_t = \frac{q_t^b - q_t^a}{q_t^b + q_t^a}. \tag{6}$$

I also calculate the top-of-book OFI increment of Cont et al. (2014),

$$\begin{aligned} e_t = \mathbf{1}_{\{P_t^b \geq P_{t-1}^b\}} q_t^b - \mathbf{1}_{\{P_t^b \leq P_{t-1}^b\}} q_{t-1}^b \\ - \mathbf{1}_{\{P_t^a \leq P_{t-1}^a\}} q_t^a + \mathbf{1}_{\{P_t^a \geq P_{t-1}^a\}} q_{t-1}^a. \end{aligned} \tag{7}$$

Let $\tau(t)$ be the state at which the current mid-price was established. Only already observed increments enter

$$E_t = \sum_{u=\tau(t)+1}^{t} e_u, \qquad Z_t = \frac{2}{\pi} \arctan\left(\frac{E_t}{q_t^b + q_t^a}\right). \tag{8}$$

Thus E_t resets after each mid-price change, while Z_t is a bounded, depth-normalized summary of submissions, cancellations, and executions since that change. It is not a reconstructed queue or an estimate of order priority.

The label is one when the next same-session mid-price change is upward and zero when it is downward. I retain states after an event changes a best price or size. C++ aggregates (I_t, Z_t) onto a fixed 31×31 grid by date and spread regime; no row-level state enters the statistical package.

The first 198 included dates through 17 October fit four logistic specifications: intercept, I_t, Z_t, and (I_t, Z_t). All 51 dates from 18 October onward are scored without refitting. The intercept-only benchmark uses the training up-move frequency. Brier improvements and intervals resample complete test dates.

Table 6. Linear next-move signal ablation on the fixed late-2019 test dates. Brier reduction is relative to the training-frequency benchmark.

Spread	Signal	Observations	ROC-AUC	Brier reduction
All	Queue	5,489,604	0.538	0.12%
All	OFI	5,489,604	0.616	5.61%
All	Queue + OFI	5,489,604	0.627	5.61%
One tick	Queue	61,829	0.627	4.70%
One tick	OFI	61,829	0.694	12.06%
One tick	Queue + OFI	61,829	0.752	18.24%

Across all spreads, OFI raises ROC-AUC from 0.538 to 0.616 and reduces Brier score by 5.61%. The combined interval is [5.36%, 5.84%]. Queue adds no detectable pooled gain over OFI: the point improvement is 0.004 percent and its complete-date interval is [−0.09%, 0.10%]. In the exploratory one-tick subset, the combined model reaches 0.752 and a 18.24% reduction [16.33%, 19.97%]. Relative to OFI alone, the reduction is 7.02% [5.22%, 8.79%].

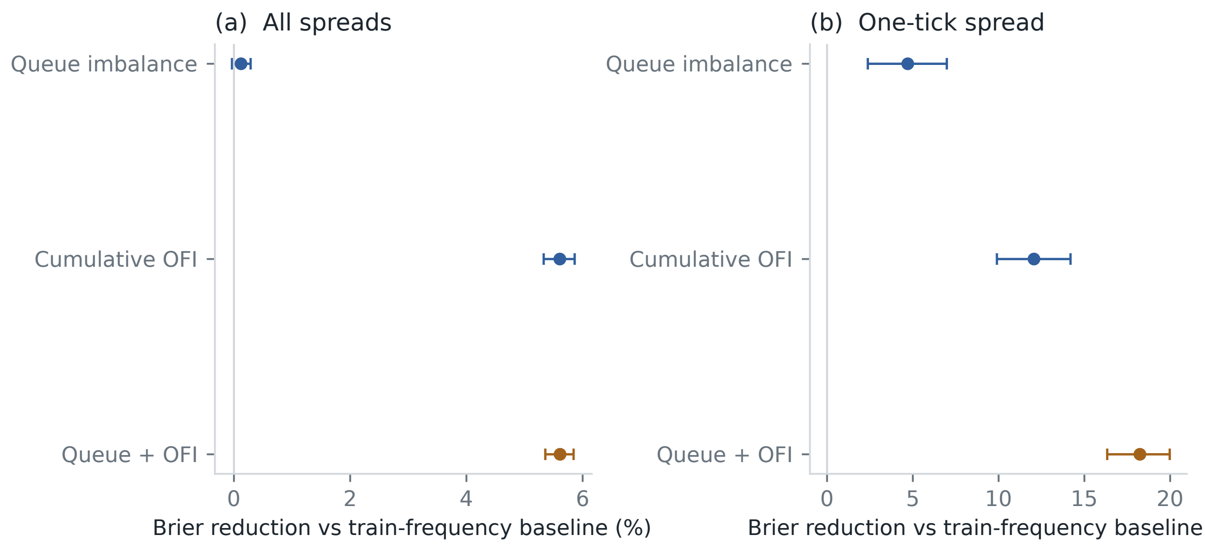

Figure 6. Relative Brier reduction against the training-frequency benchmark. Bars are 95 percent intervals from 10,000 resamples of the 51 complete test dates.

Repeating the complete ablation on fixed grids with 21, 31, and 41 bins per axis leaves the pooled combined ROC-AUC between 0.626 and 0.627 and its Brier reduction between 5.57 and

5.63 percent. The one-tick values remain 0.751–0.752 and 18.10–18.24 percent. The queue-only calibration below remains useful as a univariate diagnostic. A direction score before fees, latency, and queue position is not a trading strategy.

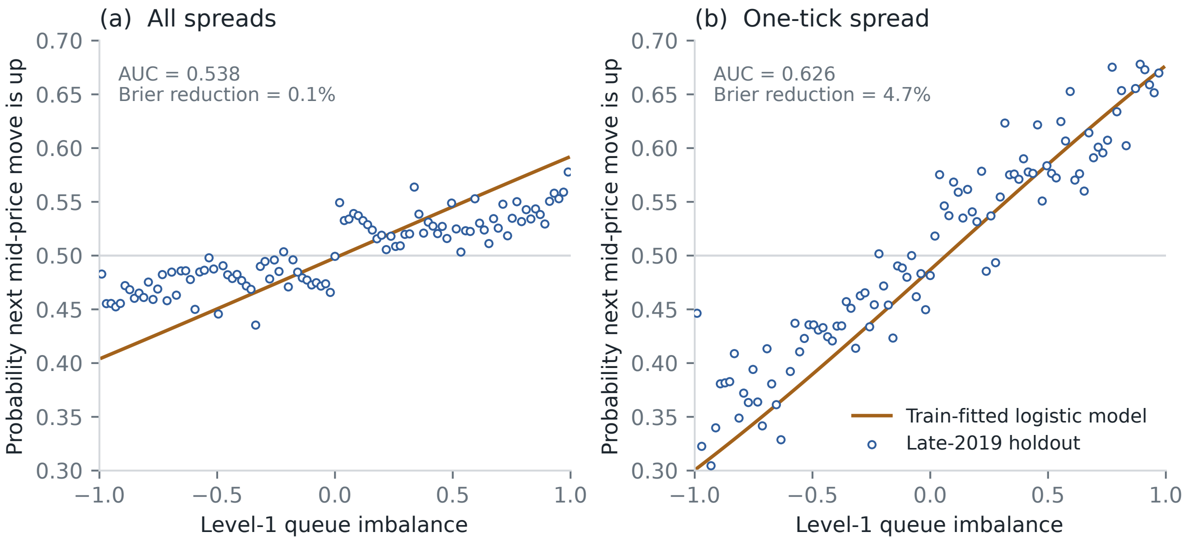

Figure 7. Next-move calibration for best-quote states. The orange curve is fitted on the first 198 included dates; blue circles are empirical frequencies on the final 51 dates. Test bins with fewer than 100 observations are omitted from the scatter but remain in the scores.

7.4 Selecting a fixed OFI lookback without test leakage

The since-price-change feature has a variable horizon: a long-lived price spell accumulates more OFI than a short-lived one. I therefore compare it with two families of fixed, causal lookbacks. If $j(t)$ indexes best-quote updates up to state t, the update-count feature sums the current and preceding $K-1$ OFI increments for $K \in \{1, 5, 20, 100\}$. The clock-time feature uses

$$E_t^{(\Delta)} = \sum_{u:\, t-\Delta < t_u \le t} e_u, \qquad \Delta \in \{10, 100, 1000, 10000\} \text{ microseconds.} \tag{9}$$

Both families can cross the preceding price-spell boundary and include the observed update at t. The adaptive reference instead discards the transition that established the current mid-price and starts accumulating immediately after it. Every candidate uses the same depth normalization, arctangent transform, 31×31 grid, labels, dates, and spread buckets.

The final 51 dates do not choose the horizon. The first 148 dates fit each fixed candidate, and the next 50 dates select the lowest Brier score separately for each spread sample and model. The selected candidate is then refit on all 198 pre-test dates and scored once on the final 51. The price-spell reset is excluded from selection and retained as the adaptive reference. Rejected fixed candidates retain no final-period scores. Complete-date bootstrap comparisons pair the two evaluated losses on the same holdout dates.

Table 7. Train-only fixed-horizon selection for the combined queue/OFI model. Validation and test reductions use the corresponding earlier-date frequency baseline. The last column is the selected fixed window's relative Brier error increase over the adaptive price-spell reset on the final 51 dates.

Spread	Fixed lookback	Val. reduction	Test AUC	Test reduction	Increase vs reset [95%]
All	10 ms	3.97%	0.613	3.64%	2.09% [1.77%, 2.43%]
One tick	1 ms	7.30%	0.715	13.57%	5.71% [3.86%, 7.65%]

The pooled validation block selects a 10 ms clock window; the one-tick block selects 1 ms. Both retain substantial directional information, but both lose to the adaptive reset on untouched dates. The relative Brier penalties are 2.09% [1.77%, 2.43%] and 5.71% [3.86%, 7.65%]. This negative result supports the original reset rule within the pre-specified candidate set; it does not establish that price-spell OFI is universally optimal.

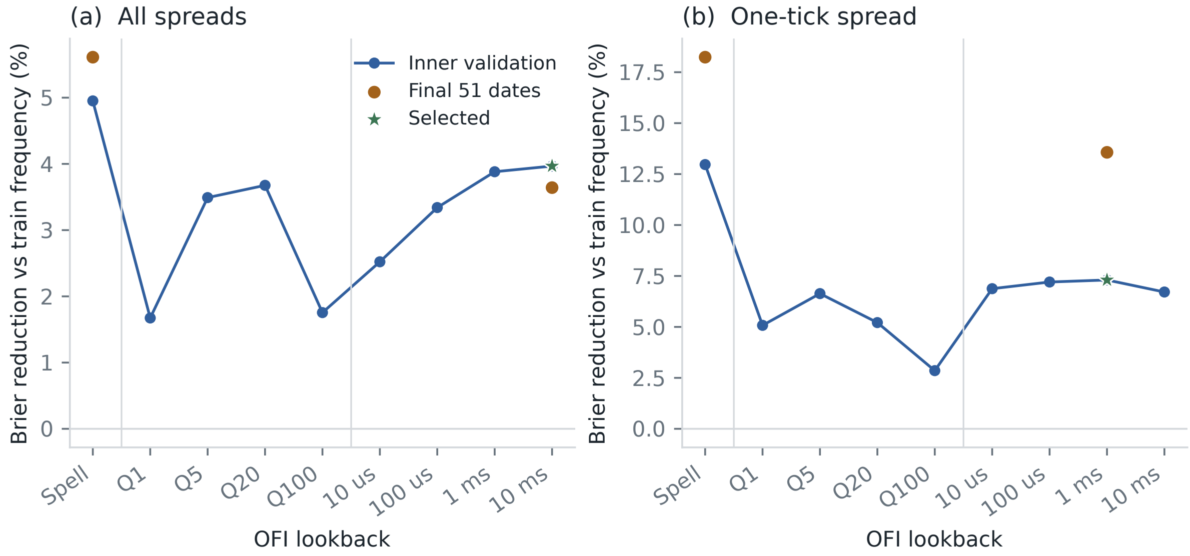

Figure 8. Combined-model Brier reductions for the adaptive price-spell reset, fixed quote-update windows (Q), and fixed clock windows. Stars mark the fixed horizon selected on the inner validation dates. Orange final-period scores appear only for that selection and the pre-specified adaptive reference.

7.5 Marketable markout gate

The direction scores do not include the cost of crossing the book. I therefore apply a separate, deliberately narrow execution diagnostic to the same best-quote-update states. For each logistic model, the first 198 dates determine confidence cutoffs targeting 100, 20, 10, and 5 percent of training signals, where confidence is $|\hat{p}_t - 0.5|$. A retained state buys when $\hat{p}_t \geq 0.5$ and sells otherwise. No holdout outcome or holdout probability quantile sets a cutoff.

Let $s_t \in \{-1, 1\}$ be that decision, let $m_{t,\delta}$ and $h_{t,\delta}$ be the prevailing midpoint and half-spread when latency δ elapses, and let $m_{\tau(t)^+}$ be the midpoint immediately after the next price change. At positive latency, a next move stamped at or before the entry deadline makes the signal stale. Otherwise,

$$G_{t,\delta} = 10^4 s_t \frac{m_{\tau(t)^+} - m_{t,\delta}}{m_{t,\delta}}, \tag{10}$$

$$N_{t,\delta} = G_{t,\delta} - 10^4 \frac{h_{t,\delta}}{m_{t,\delta}}. \tag{11}$$

G is the signed midpoint markout and N subtracts the displayed half-spread. The exporter evaluates $\delta \in \{0, 10, 100, 1000, 10000\}$ microseconds. Entry uses the last displayed book state strictly before a positive-latency deadline, so an event stamped exactly at that deadline is excluded. Event ordering defines the zero-latency post-event state.

Table 8 reports the combined queue/OFI model. Across all spreads, the 10-percent rule has a zero-latency gross markout of 0.312 basis points but pays 1.565 basis points of half-spread, leaving -1.254 $[-1.364, -1.156]$. Directional information is therefore too small to cover marketable entry in the pooled sample.

Table 8. Combined-model marketable markouts on the final 51 dates. Confidence fractions and model coefficients are fixed on the first 198 dates. Intervals are from 10,000 complete-date bootstrap resamples.

Train-confidence	Spread	Latency	Executable	Stale	Net bp [95% interval]
10%	All	0 us	555,039	0.0%	-1.254 [-1.364, -1.156]
10%	One tick	0 us	5,787	0.0%	0.399 [0.298, 0.500]
10%	One tick	10 us	3,367	41.8%	0.302 [0.135, 0.453]
10%	One tick	100 us	1,311	77.3%	-0.035 [-0.422, 0.330]
5%	One tick	100 us	482	84.0%	0.167 [0.075, 0.267]
5%	One tick	1 ms	195	93.5%	0.013 [-0.086, 0.133]

The exploratory one-tick subset behaves differently. At the 10-percent rule, the immediate midpoint markout is 0.545 basis points and the half-spread is 0.146, leaving 0.399 [0.298, 0.500]. At 10 microseconds, the net remains 0.302 [0.135, 0.453], although 41.8% of selected states have already reached their next mid-price move. At 100 microseconds, 77.3% are stale and the 10-percent interval crosses zero.

The pre-specified 5-percent tail remains positive at 100 microseconds, but only 482 holdout states are executable; by 1 millisecond its interval crosses zero. Model ordering also changes with the cost diagnostic: at the 10-percent rule and 100 microseconds, OFI alone has a net markout of 0.102 basis points [0.010, 0.203], while the combined interval crosses zero. Higher next-move Brier improvement does not automatically imply a better execution-conditioned tail.

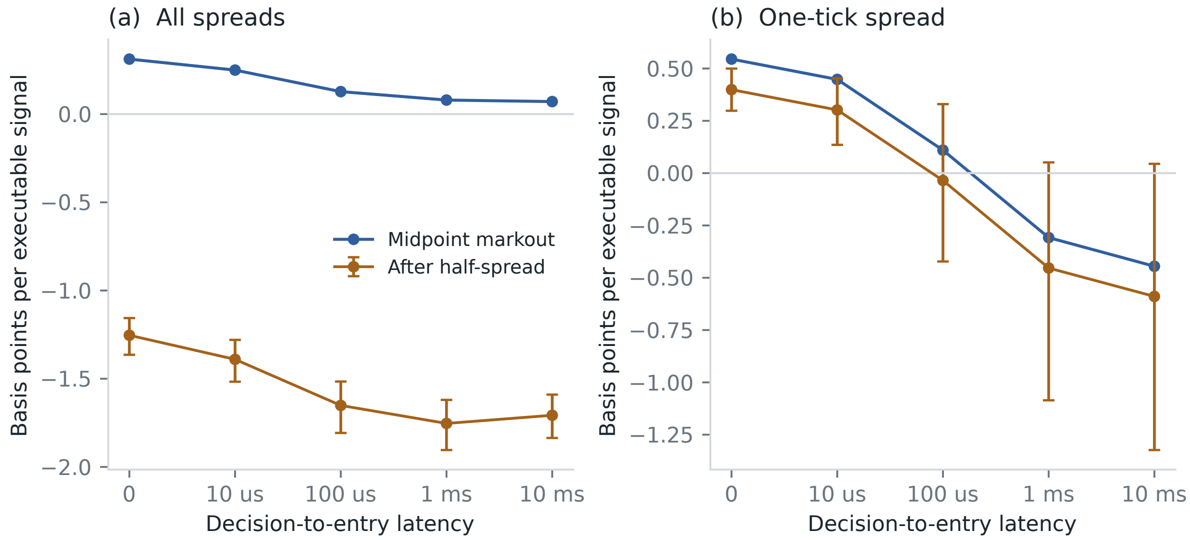

Figure 9. Gross midpoint and half-spread-adjusted markouts for the combined model's 10-percent training-confidence rule. Orange bars are 95 percent intervals from complete-date resampling. Signals whose next price move precedes entry are excluded from the per-execution mean and reported as stale in Table 8.

This gate is not a backtest. Several states in one constant-mid-price spell share the same terminal mark, so their hypothetical trades are not additive. The order size is one unit, displayed depth is assumed fillable, and fees, market impact, inventory, capital, and risk limits are absent. The imposed delay is applied to LOBSTER event time without separating feed, computation, transmission, or exchange processing. The one-tick stratum, confidence tails, and latency comparisons are multiple exploratory views without a family-wise correction.

7.6 One clock-time landmark per price spell

The preceding gate deliberately evaluates every qualifying best-quote update. I test the sensitivity to repeated labels with a second protocol that pre-specifies one observation time per constant-mid-price spell. If t_r is the event that establishes spell r, the landmark is $\ell_r = t_r + 100$ microseconds. A spell ending at or before ℓ_r produces no signal. Otherwise, queue imbalance, cumulative OFI, and the spread use the last book state strictly before ℓ_r. Events stamped exactly at the deadline remain unobserved, so same-timestamp ordering cannot leak into the signal. Entry latency is then applied after ℓ_r.

Each eligible spell contributes at most one decision, and successive spells cannot overlap before their terminal price changes. Of 15,263,228 completed spells in the included-session replay, 9,847,952 (64.5%) survive to the fixed landmark. The first 198 dates again fit the linear queue, OFI, and combined models; the final 51 dates remain untouched. Table 9 separates direction quality from the later cost calculation.

Table 9. Combined-model direction scores at the 100-microsecond price-spell landmark. Brier reductions compare with the training-frequency intercept; intervals resample complete test dates.

Spread	Test landmarks	ROC-AUC	Brier reduction [95% interval]
All	2,198,301	0.555	0.91% [0.83%, 0.98%]
One tick	8,723	0.650	7.00% [5.65%, 8.45%]

The fixed grid creates probability ties. The cutoff targeting 10 percent of training signals therefore retains 13.3% across all spreads and 13.5% in the one-tick stratum; no tie is broken using test outcomes. The pooled markout remains decisively negative. In the exploratory one-tick stratum, the immediate interval is positive but is much smaller than in the repeated-state diagnostic. It remains just above zero after 10 microseconds and is negative after 100 microseconds.

Table 10. Combined-model markouts from the non-overlapping 100-microsecond landmarks on the final 51 dates. The rule targets 10 percent train coverage; intervals use 10,000 complete-date bootstrap resamples.

Spread	Entry latency	Executable	Stale	Net bp [95% interval]
All	0 us	248,800	0.0%	-1.909 [-2.074, -1.752]
One tick	0 us	1,120	0.0%	0.107 [0.045, 0.179]
One tick	10 us	1,006	10.2%	0.059 [0.001, 0.125]
One tick	100 us	689	38.5%	-0.095 [-0.155, -0.035]

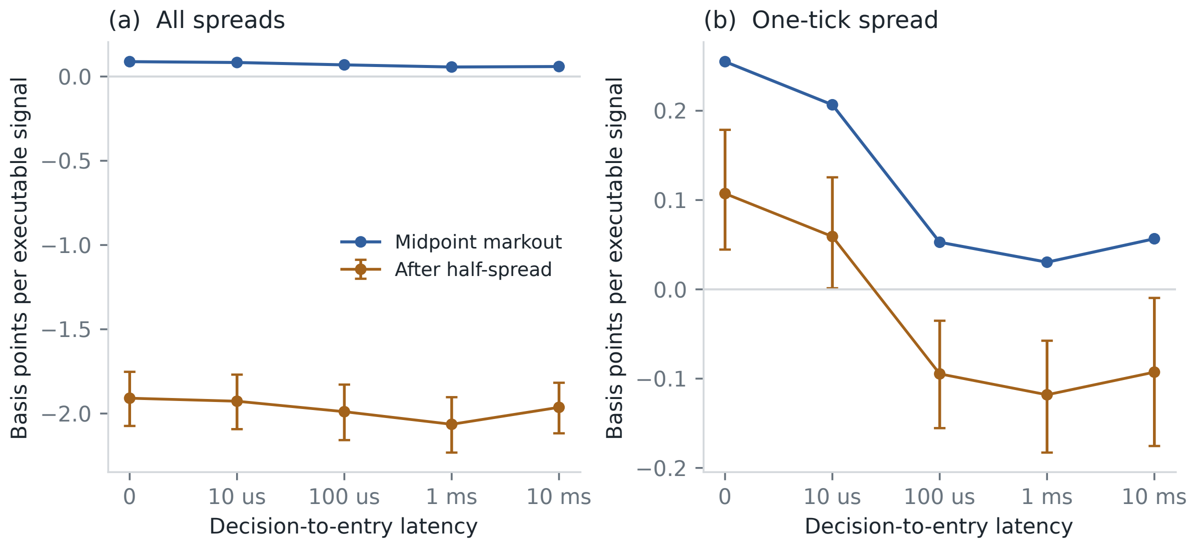

Figure 10. Gross and half-spread-adjusted markouts after the fixed 100-microsecond observation landmark. Each eligible price spell contributes at most one signal. Orange bars are complete-date 95 percent intervals.

Removing within-spell repetition changes both the sample and the fitted confidence cutoff, so the two markout gates are not paired estimators. The attenuation nevertheless shows that the positive one-tick tail is sensitive to the decision clock. Non-overlap also does not turn the calculation into a backtest: the terminal midpoint is a mark rather than an executable exit, and fees, impact, fill uncertainty, inventory, capital, and risk limits remain absent.

7.7 Rolling-origin stability of the landmark signal

The preceding direction result still depends on one 198/51-date origin. I therefore add a split-sensitivity audit following the rolling-origin principle described by Tashman (2000). January through June form the initial estimation period. At the start of each month from July through December, the same intercept, queue, OFI, and combined specifications are refit on every preceding date; that month's landmarks are then scored once. No later date enters an earlier fit, and no model or feature is selected from the six evaluation months.

For model k, let $L_{d,k}$ be the aggregate Brier loss on date d, calculated directly from the up- and down-move counts in each fixed signal cell. The six-month comparison with reference model 0 is

$$\Delta_{k,0} = 1 - \frac{\sum_d L_{d,k}}{\sum_d L_{d,0}}. \tag{12}$$

The primary comparison is the combined model against the prior-date up-move frequency across all spreads. To retain short-range dependence between trading dates, I resample the paired daily loss sequences with the circular-block procedure of Politis and Romano (1992). The reported interval uses five-date blocks and 10,000 draws; one- and ten-date blocks are retained as sensitivity checks.

Table 11. Monthly rolling-origin direction audit on the non-overlapping landmarks. Each row combines 127 dates from July through December. Intervals use paired circular blocks of five consecutive trading dates.

Spread	Landmarks	ROC-AUC	Brier reduction [95%]	Positive months
All	4,904,599	0.562	1.17% [0.93%, 1.45%]	6 / 6
One tick	32,651	0.635	5.55% [4.24%, 6.62%]	6 / 6

The combined reduction is positive in every month for both samples. Relative to OFI alone, it is 0.54% [0.32%, 0.80%] across all spreads and 3.21% [1.67%, 4.51%] in the exploratory one-tick subset. That secondary comparison is positive in six of six pooled months and five of six one-tick months. Expanding the circular blocks to ten dates leaves lower bounds of 0.87% and 4.06% against the frequency benchmark.

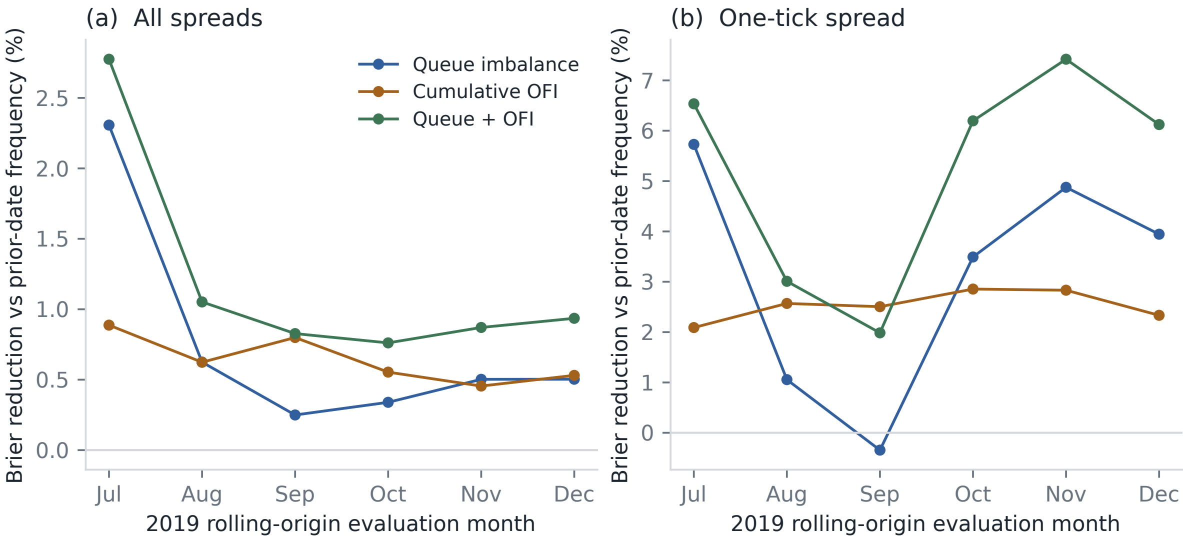

Figure 11. Monthly rolling-origin Brier reductions at the fixed price-spell landmark. Each model is refit only on dates preceding the displayed month. The one-tick sample remains exploratory.

This is not a second untouched holdout. The protocol was specified after the fixed 198/51 result was inspected, and a month enters the expanding training set after it has been scored. The audit shows that the direction result is not confined to one terminal split and is not erased by one- or two-week loss blocks. It does not account for feature-search history, replace validation on another stock-year, or alter the negative quoted-execution conclusion below.

7.8 Depth-constrained quoted round trips

I next replace the terminal midpoint with a second marketable order. This is a stress test of the preceding result, specified after that midpoint result had already been inspected, rather than a new untouched validation. The signal, 100-microsecond landmark, train/test split, model, and confidence cutoff remain unchanged.

For order size q, a long decision buys the displayed asks at entry and sells the displayed bids in the first snapshot after the next mid-price change. A short decision uses the reverse path. Each leg walks at most the two observed levels. If either endpoint cannot display all q shares, that round trip is capacity-censored rather than extrapolated into unobserved depth. Within each filled aggregate cell, the quoted result obeys

$$P^{\text{quoted}} = P^{\text{midpoint}} - C^{\text{entry}} - C^{\text{exit}}.$$

Both possible action paths are aggregated before Python applies the train-fitted direction rule, so no row-level outcome or test-selected direction enters the model.

The decomposition makes the earlier one-tick result transparent. At one share and zero entry latency, the signed midpoint move is 0.255 basis points and entry costs 0.148, reproducing the earlier 0.107 basis-point markout after rounding. Crossing the post-move book costs another 0.791 basis points. The quoted round trip is therefore -0.684 $[-0.756, -0.619]$. Across all spreads, the corresponding decomposition is 0.088 minus 1.997 minus 2.032, or -3.941 basis points.

Table 12. Depth-constrained quoted round trips for the combined model's 10-percent training-confidence rule on the final 51 dates. Capacity is the fraction of arrived signals lacking the requested displayed size on at least one leg. Intervals resample complete dates.

Spread	Entry latency	Shares	Filled	Capacity	Zero-fee PnL [95% interval]
All	0 us	1	248,800	0.0%	-3.941 [-4.274, -3.624]
All	0 us	100	131,090	47.3%	-5.174 [-5.573, -4.779]
One tick	0 us	1	1,120	0.0%	-0.684 [-0.756, -0.619]
One tick	10 us	1	1,006	0.0%	-0.713 [-0.794, -0.641]
One tick	100 us	1	689	0.0%	-0.790 [-0.892, -0.703]
One tick	0 us	100	532	52.5%	-1.398 [-1.546, -1.259]
One tick	0 us	500	25	97.8%	-1.891 [-3.097, -1.104]

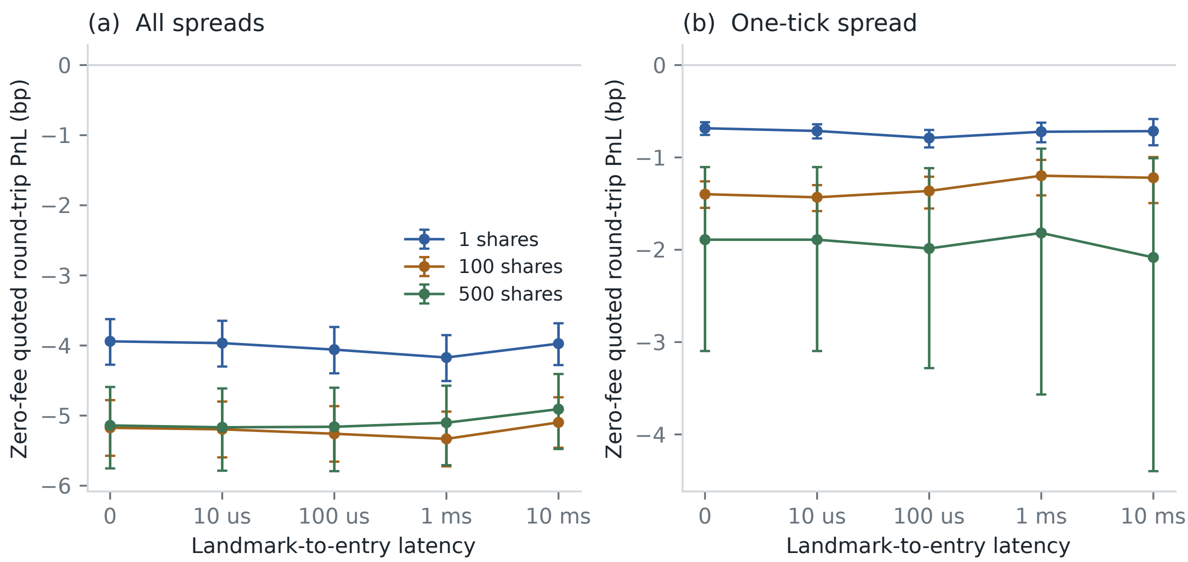

Figure 12. Zero-fee quoted round trips after the fixed price-spell landmark. Both entry and exit walk up to two displayed levels. Bars are complete-date 95 percent intervals; missing tail points have no fully displayed round trip.

The result is negative before any fee for every displayed scenario in Table 12. Increasing size also changes the evaluated sample: at 100 shares, 52.5% of arrived one-tick signals are capacity-censored; at 500 shares, only 25 round trips remain. These are not capacity estimates for a strategy. The committed 360-row grid retains every specified scenario, including the three without a fill; their economic fields remain null.

7.8.1 Selection-aware policy-family audit

The complete output crosses spread sample, direction model, training-confidence target, entry latency, and order size into 360 policies. Choosing the least negative row after viewing that grid invalidates its pointwise interval. The audit therefore reconstructs each policy from the aggregate cells and stored train-only parameters and stops unless its selected, arrived, filled, and quoted-PnL totals match exactly.

For policy j and test date d, let P_{dj} be aggregate quoted PnL and F_{dj} its fill count. The ratio estimate is $\widehat{\mu}_j = \sum_d P_{dj} / \sum_d F_{dj}$, with influence numerator $P_{dj} - \widehat{\mu}_j F_{dj}$. Ten thousand common Rademacher multiplier draws over the 51 dates preserve cross-policy dependence; their maximum absolute studentized statistic forms simultaneous 95 percent intervals (Chernozhukov et al., 2013), with critical value 3.026. 357 policies fill; 356 fill on at least two dates with nonzero cluster variance. Three zero-fill policies and one single-date policy remain explicit without artificial intervals.

The test-set winner across all 360 rows is the one-tick OFI model at 5 percent target coverage, zero latency, and one share. Its 565 fills span all 51 dates and average -0.615 basis points. Its pointwise interval is $[-0.682, -0.561]$ and its family-wide interval is $[-0.709, -0.520]$, with $p_{\text{adj}} < 0.0001$.

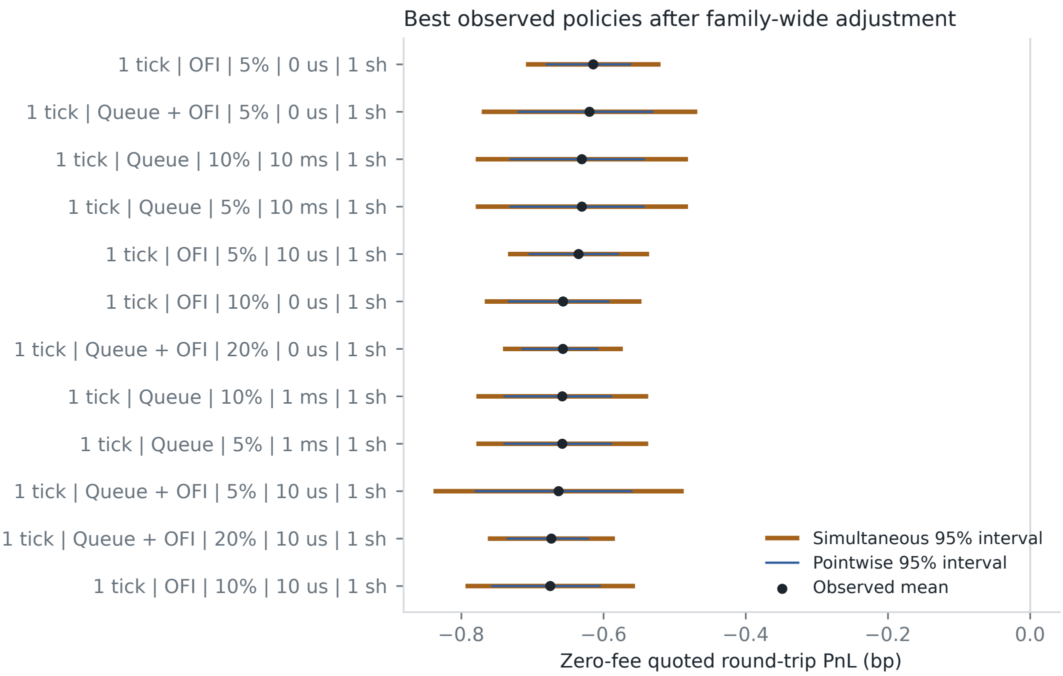

Figure 13. Pointwise and simultaneous intervals for the twelve highest observed quoted-PnL policies. Ranking uses the final dates; the orange intervals share one multiplier distribution over all 356 estimable policies.

332 of 356 simultaneous upper bounds are below zero. Sparse, mainly large-size tails prevent a claim that every policy is negative. The selection-valid conclusion is narrower: the best observed configuration remains negative under this quoted execution protocol.

The diagnostic is deliberately optimistic in other respects. Exit uses the first snapshot after the next price move with zero reaction latency, displayed size is assumed simultaneously available to the hypothetical order, and no fee is charged. Any non-negative fee lowers the reported PnL. Market impact beyond walking the two displayed levels, queue competition, inventory, capital,

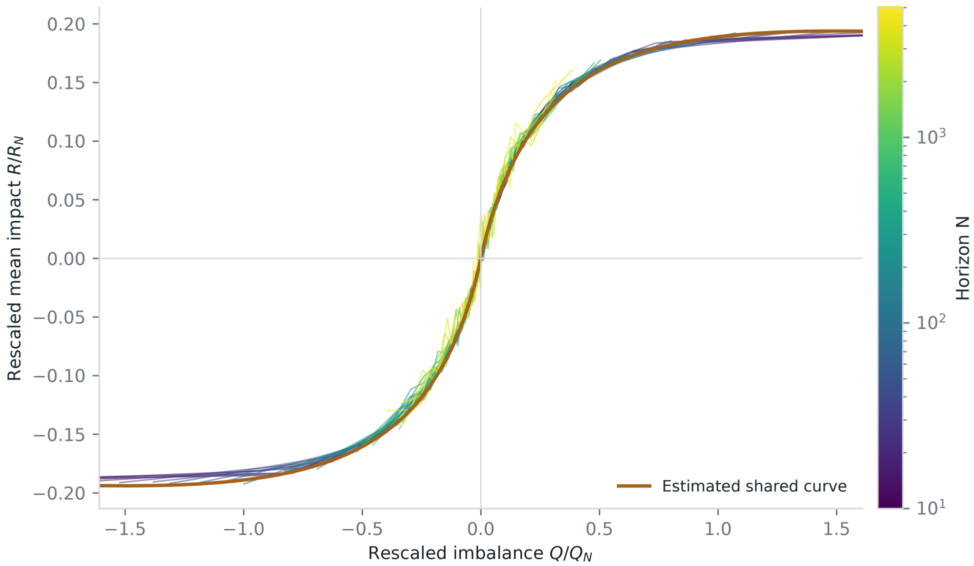

Figure 14. Finite-size scaling collapse for TSLA 2019. Color identifies the aggregation horizon. The orange line is the common fitted shape. This is an in-sample curve construction, not a forecast evaluation.

and risk limits remain absent. The negative result rejects an economic reading of the positive midpoint markout under this quoted execution protocol; it does not estimate deployable returns.

8 Scaling analysis

8.1 Construction

The scaling exercise follows the aggregate-volume form in Patzelt and Bouchaud (2018). For a window of N timestamp-aggregated type-4 and type-5 transactions, let Q be signed share volume after a daily-volume adjustment, and let r be the log mid-price change. Conditional curves are estimated in 31 quantile bins for 28 log-spaced horizons from 10 to 5,000 transactions.

Each curve is modeled as

$$\mathcal{R}_N(Q) = R_N F(Q/Q_N), \tag{13}$$

$$F(x) = \frac{x}{(1+|x|^{\alpha})^{\beta/\alpha}}. \tag{14}$$

One width Q_N and height R_N are estimated per horizon, with a common α and β. The nonlinear fit uses observation-count weights and a soft-L1 loss. Figure 14 plots the resulting rescaled curves. The collapse is close over the central observed range, with more dispersion around the origin and at a few long-horizon curves.

8.2 Scale fits and interpretation

After the curve fit, the scale parameters are regressed on horizon in log space:

$$Q_N = a_Q N^{\xi}, \qquad R_N = a_R N^{\psi}. \tag{15}$$

Table 13 and Figure 15 report those second-stage fits. The width exponent is 0.976 and the height exponent is 0.656. The residual panels show that the largest horizons carry the main deviations from a straight power law.

Table 13. Log-log fits to the 28 estimated scale parameters. Standard errors condition on the first-stage scale estimates.

Scale	Exponent	Standard error	In-sample R^2
Width Q_N	0.976	0.012	0.9958
Height R_N	0.656	0.006	0.9975

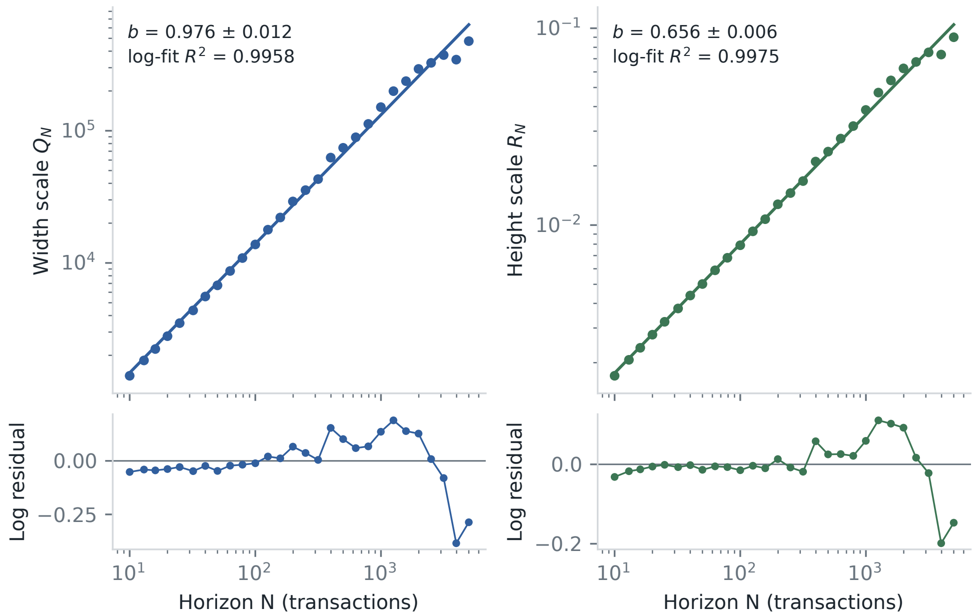

Figure 15. Power-law fits for the estimated width and height scales. Lower panels show log residuals. The displayed R^2 values describe these 28 in-sample points only.

Near zero imbalance, the master-curve slope scales as R_N/Q_N. The fitted exponents imply a decay $N^{-(\xi-\psi)}$ with $\xi - \psi = 0.320$. The local Kyle slopes in Section 4 decay with exponent 0.315. The two calculations use different transaction samples and return definitions, so their agreement is descriptive.

The pooled order-sign standard deviation scales with horizon exponent 0.693. This is a descriptive variance-scaling fit, not a DFA estimate or sufficient evidence of long memory. The 0.9958 and 0.9975 values belong to regressions on 28 estimated scale points. They do not measure future price variation. The proposed scaling form describes this included TSLA sample in-sample; the cross-asset claim in Patzelt and Bouchaud (2018) requires a broader panel.

8.3 Daily order-sign persistence

The pooled exponent above mixes all dates and does not test whether the sign sequence is persistent within a session. I therefore apply first-order detrended fluctuation analysis (Peng et al., 1994) separately to every date. For each sign sequence, I subtract its daily mean, integrate the centered signs, split the profile into non-overlapping segments from both the beginning and end, and remove a linear trend in every segment. The scale grid was fixed at 16, 32, 64, 128, 256, 512, and 1,024 transactions. The exponent is the equal-weight slope through the seven log fluctuation values.

A date enters when it contains at least 8,192 signed transactions. All 249 included sessions pass this rule, covering 100.0% of the 5,845,437 type-4/type-5 transactions. No date is filtered by its exponent, fit quality, or curve shape. The daily median fit R^2 is 0.9957.

Table 14. Daily DFA1 order-sign audit. The null distribution contains 199 annual medians from independent within-date permutations. Date intervals use 10,000 circular-block bootstrap draws.

Quantity	Estimate
Daily median exponent	0.699
Daily interquartile range	[0.681, 0.720]
Permutation-null median	0.501 [0.498, 0.503]
Upper-tail Monte Carlo p-value	0.005
One-date block interval	[0.696, 0.705]
Five-date block interval	[0.694, 0.708]
Ten-date block interval	[0.694, 0.709]

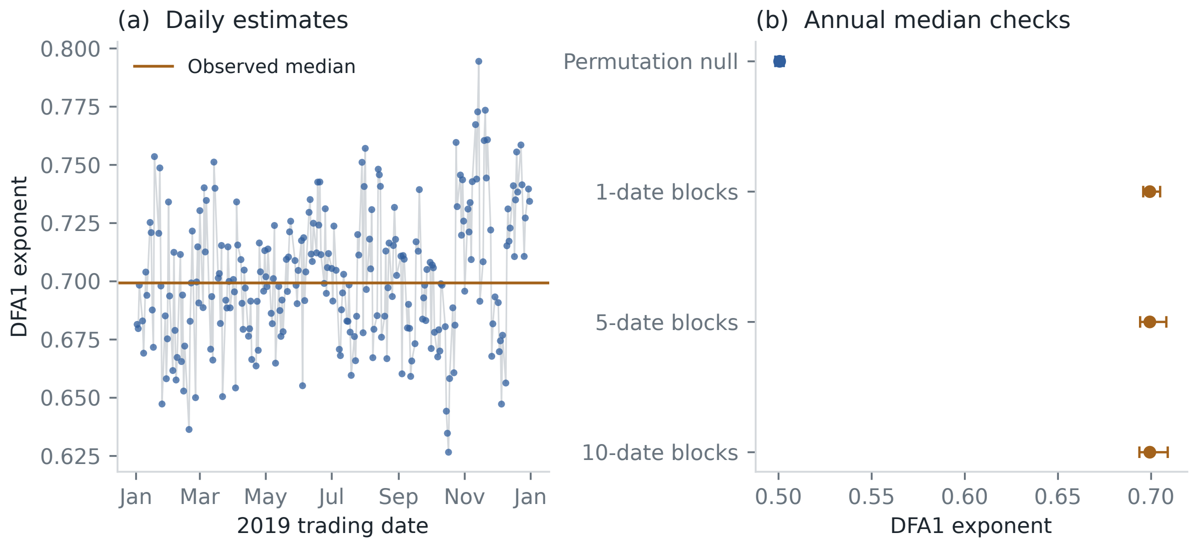

Figure 16. Daily DFA1 estimates and annual-median checks. The left line is the observed annual median. On the right, the blue interval is the central 95 percent range of the permutation-null medians; orange intervals resample consecutive trading dates. The two interval types answer different questions.

None of the 199 permuted annual medians reaches the observed value. The plus-one Monte Carlo estimate is therefore $p_{\mathrm{MC}} = 0.005$. The null preserves each date's transaction count and buy fraction, so this rejects an independently ordered sign sequence under that conditioning. It demonstrates serial dependence over the fixed scales. It does not by itself identify asymptotic long memory: DFA1 does not remove every intraday seasonal pattern or structural break, and only one stock-year is available.

9 Predictability-conditioned liquidity

The DFA result establishes dependence but does not explain how comparatively weak return dependence can coexist with predictable flow. Following the diagnostic in Lillo and Farmer (2004), I fit one linear sign predictor,

$$\widehat{\epsilon}_t = a_0 + \sum_{k=1}^{50} a_k \epsilon_{t-k}. \tag{16}$$

Lags reset at every session boundary. The coefficients are estimated once by OLS on the first 198 dates; the first 50 orders of each date are omitted. This gives 3,512,254 training observations and 1,022,374 observations on the fixed final 51 dates. The reference is the constant mean sign estimated on the same training rows.

For each realized side, define expectedness as $z_t = \epsilon_t \widehat{\epsilon}_t$. Ten quantile boundaries are estimated on the training dates separately for buys and sells, then held fixed. Let V_t be the reconstructed visible-order size and Q_t^{opp} the opposite best-quote size immediately before its first fill. The penetration proxy and relative-liquidity outcome are

$$\mathbf{1}\{V_t \geq Q_t^{\mathrm{opp}}\}, \qquad \log(V_t/Q_t^{\mathrm{opp}}). \tag{17}$$

The first is a penetration proxy: reconstructed size meets or exceeds the initial best quote. It is not total depth, a direct observation of depletion, or executable capacity. Contrasts compare the most expected and most surprising train-defined bins, giving buys and sells equal weight. I report $\log V_t$ and $\log Q_t^{\mathrm{opp}}$ separately because their difference is exactly $\log(V_t/Q_t^{\mathrm{opp}})$.

Displayed depth varies through the day, so a post-hoc sensitivity compares the two tails within fixed strata

$$h = (\mathrm{date}, \mathrm{side}, \lfloor(\text{seconds after 09:30})/1800\rfloor). \tag{18}$$

Within each stratum containing both tails, the difference receives weight $n_{h,0}n_{h,9}/(n_{h,0} + n_{h,9})$. This is the within-stratum least-squares weight. I first pool those sufficient statistics for each side, then average the buy and sell estimates. The control retains 218,905 / 222,243 tail observations (98.5%) in 1,235 / 1,302 half-hour strata with common support.

Table 15. Fixed later-date sign forecast and conditional liquidity. Intervals use 10,000 circular bootstrap draws with five-date blocks.

Quantity	Estimate
Sign accuracy	80.9%
Sign MSE reduction vs training-mean reference	39.6% [38.1%, 41.1%]
Penetration proxy, most surprising bin	63.0%
Penetration proxy, most expected bin	51.7%
Expected minus surprising proxy	-11.4 percentage points [-13.7, -8.7]
Raw mean log order-size contrast	0.099 [0.055, 0.145]
Raw mean log opposite-depth contrast	0.548 [0.468, 0.616]
Raw mean log size/depth contrast	-0.449 [-0.515, -0.374]
Intraday-adjusted penetration proxy	-10.7 pp [-13.1, -7.9]
Adjusted mean log order-size contrast	0.063 [0.021, 0.102]
Adjusted mean log opposite-depth contrast	0.494 [0.422, 0.552]
Adjusted mean log size/depth contrast	-0.432 [-0.499, -0.357]

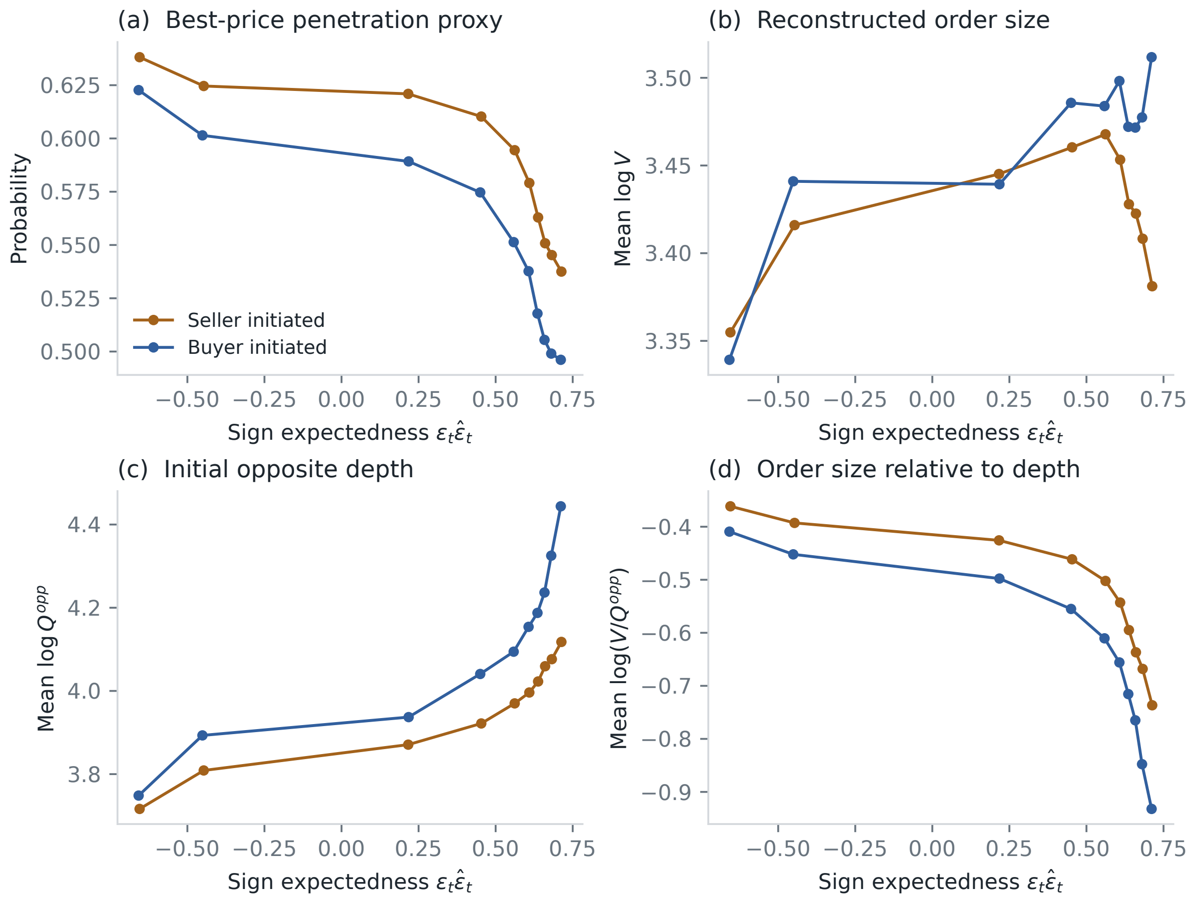

Figure 17. Liquidity conditional on sign expectedness in the final 51 dates. Bin boundaries are learned on the earlier dates separately by realized side. The four panels separate penetration, order size, opposite depth, and their log ratio. They show an association, not a causal response by liquidity providers.

The AR(50) clears its mechanism gate: later-date sign MSE falls by 39.6%, with interval [38.1%, 41.1%]. At the same time, the probability of meeting or exceeding the initial opposite best declines from 63.0% in the most surprising bin to 51.7% in the most expected bin. The raw log size/depth contrast is -0.449. Its components rule out the simple explanation that expected orders are smaller: their mean log size is 0.099 higher, while mean log opposite depth is 0.548 higher.

The half-hour control gives the same attribution. Mean log order size is 0.063 higher, but mean log opposite depth is 0.494 higher, leaving a -0.432 log-ratio contrast. The adjusted penetration difference is -10.7 pp. One- and ten-date blocks preserve the signs of all four adjusted contrasts. These intervals condition on the earlier-date fit and do not cover model choice or a new market regime.

This pattern is compatible with the asymmetric-liquidity account in Lillo and Farmer (2004); Farmer et al. (2006): the orders that are easiest to anticipate face more displayed depth relative to their size. The decomposition identifies which observed component dominates, but not who changed the book, when they did so, or whether the response makes prices efficient. Timestamp-and-side grouping is a proxy for a parent aggressive order, not an investor identifier. The split is chronological for this model, but the final dates had already been examined by other project analyses, so this is not a globally untouched replication.

10 Limitations

- The variables and response cover the same window, and order flow is endogenous. The fitted coefficients are neither pre-trade forecasts nor causal price effects.
- The replay checks finite-depth state transitions against authoritative snapshots. It cannot recover orders submitted outside the requested depth, FIFO priority, network timing, or exchange matching logic.
- Timestamp and side grouping cannot recover investor identity or institutional metaorders. The book covers NASDAQ rather than consolidated US liquidity, and hidden-execution signs are uncertain around the midpoint.
- The predictability-conditioned audit uses realized trade side to select opposite depth and standardize expectedness bins. That depth is an outcome measured against a strictly lagged sign forecast, not a predictor of the current sign. Initial best size is neither total book depth nor fillable capacity. The fixed half-hour control removes date, side, and coarse intraday composition from the tail contrast, but not spread regimes or unobserved trader decisions. The association does not identify a causal market-efficiency mechanism.
- Several states in the main next-move experiment can share one eventual label. The 100-microsecond landmark removes that repetition for a separate protocol, but filters short spells and changes the target population. Uncertainty is resampled at the trading-date level. The fixed-horizon experiment tests eight alternatives, but they include the update that establishes a new mid-price and can cross spell boundaries. It therefore does not isolate horizon length from reset semantics or exhaust all causal OFI summaries.
- The rolling-origin audit reduces dependence on one terminal split but was specified after that split was inspected and reuses the same stock-year. Its circular-block intervals condition on the fitted forecast sequence and probe only three block lengths; they do not account for feature-search history, nonstationarity across years, or model-estimation uncertainty in a new market regime.
- The two midpoint-markout gates assume a one-unit fill at the displayed best and do not provide an executable exit. The round-trip extension instead walks the displayed level-2 book, but assumes that liquidity is simultaneously available and exits with zero reaction latency. Fees, market impact beyond two levels, queue competition, inventory, capital, and risk limits remain absent. The stated entry latency is one exogenous delay applied to LOBSTER event time; it does not identify components of an implementable system.
- The earlier state-level and midpoint-markout grids remain exploratory and receive no family-wise correction. The quoted round-trip audit provides simultaneous coverage for 356 estimable members of its 360-policy family, but it was added after the midpoint result and does not account for the broader feature and protocol research process.
- The one-tick spread stratification is exploratory. Its stronger result is not evidence of generalization beyond this stock-year.
- The joint logistic models use fixed binned features rather than raw event states. The 21/31/41-grid check limits this discretization concern but does not replace validation on an untouched stock-year.
- The feature family grew out of the course analysis. Although dates are held out chronologically, there is no second untouched year after the feature comparison. The bootstrap does not account for that research process.

- Three delivered source files end before their scheduled close, and replacement files are unavailable under the original project access. The canonical rebuild declares them as source exclusions and uses the other 249 sessions. This source limitation prevents a claim of complete 2019 coverage. The calendar boundaries are fixed rather than recomputed after exclusion.
- The scaling standard errors treat first-stage Q_N and R_N estimates as data. They do not propagate all curve-fitting uncertainty, and overlapping information across horizons remains.
- The daily DFA1 permutation test holds session length and buy fraction fixed but does not remove every intraday seasonal pattern, structural break, or change in execution-sign classification. Its rejection establishes serial dependence on the declared scales, not asymptotic long memory.
- TSLA in 2019 is one liquidity regime. The count, OFI, and fitted-scaling results cannot be assumed to hold for another asset or year.

11 Reproducibility and provenance

The Python package first loads the shared source and calendar policy and writes an aggregate scheduled-close audit. It then reconstructs the two transaction tables from the included sessions, runs the daily sign-persistence audit, builds event-time windows, fits the models, runs the predictable-flow liquidity audit, and writes the remaining aggregate files under `results/`. The C++20 library loads the same policy, rejects undeclared incomplete sessions, audits finite-depth transitions, and exports daily queue, joint queue/OFI, multi-horizon OFI, state-level markout, price-spell landmark, and depth-constrained round-trip bins. The figures are vector PDFs. Synthetic tests cover file pairing, source coverage, half-day filtering, book alignment, day boundaries, fixed chronological splits, complete-date and circular-block resampling, direct DFA1 detrending, queue/OFI ablation, nested train-only horizon selection, monthly rolling-origin refits, markout accounting, non-overlapping price-spell landmarks, pre-fill opposite-depth alignment, sign-predictability and intraday tail contrasts, round-trip capacity censoring, simultaneous policy inference, grid robustness, and the scaling regressions. Reproducing the annual transaction tables, replay, and aggregate grids requires licensed LOBSTER files, which are not tracked.

The course presentation lists L. Cohen, R. Darwish, A. Pariente, H. Rohrbach, and R. Sithisak. Anastasia Bugaenko supervised the work. I wrote this standalone package and report after the group project; the collaborative repository remains separate.

12 Conclusion

On the included-session late-2019 holdout, adding raw signed count to the nonlinear volume model moves R^2 from 0.298 to 0.359 and reduces MSE by 8.7%, with a date-bootstrap interval of [8.0%, 9.5%]. The full count-transform model's 27.4% reduction relative to linear signed volume is a broader ablation gap and is not attributed to count alone.

A fixed AR(50) reduces sign MSE by 39.6% on the same final 51 dates. Orders in its most expected bin meet or exceed the initial opposite best depth 51.7% of the time, versus 63.0% in the most surprising bin. Within date, side, and half-hour cells, mean log opposite depth rises by 0.494 while mean log order size rises by only 0.063, leaving a -0.432 relative-liquidity contrast. This decomposition is compatible with asymmetric liquidity, but does not establish causal adaptation or market efficiency.

The finite-depth engine audits 38,516,432 included events without claiming unobservable FIFO state. On later dates, causal OFI reduces pooled Brier score by 5.61%, while queue and OFI reach ROC-AUC 0.752 in the exploratory one-tick subset. The selected fixed lookbacks have 2.09% and 5.71% more Brier error than the adaptive reset in the pooled and one-tick samples.

Execution costs reverse the interpretation. The all-spread markout is -1.254 basis points at zero latency. The exploratory one-tick tail falls from 0.399 immediately to 0.302 at 10 microseconds; its 100-microsecond interval crosses zero with 77.3% stale. Sampling one signal per price spell leaves an all-spread markout of -1.909 and attenuates the one-tick tail from 0.107 to 0.059 and then -0.095. The rolling-origin reductions stay positive in all six later months, but only within TSLA 2019. A quoted exit then produces -0.684 basis points $[-0.756, -0.619]$ at one tick. Even the best observed row among 360 policies has simultaneous interval $[-0.709, -0.520]$. Two-level depth, zero exit reaction latency, and omitted fees, queue competition, impact, inventory, and capital make this a stress test rather than a backtest or deployable return.

The included TSLA sessions also fit the scaling form proposed by Patzelt and Bouchaud (2018) in-sample. Daily DFA1 separates the sign-persistence claim from the pooled variance fit: its median exponent is 0.699, compared with 0.501 under within-date sign permutations. The rejection supports serial dependence on the fixed transaction scales, not a claim of asymptotic long memory. None of these results is an external replication or a deployable trading strategy.

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